

Calibrated DistilBERT for Reliable Extractive QA

Project proposal | Default project, PCE division | Independent course study

Project overview

This project studies a practical weakness of extractive question-answering systems: a model can locate plausible spans even when the passage does not support an answer. The central question is whether resource-efficient SQuAD 2.0 fine-tuning, followed by held-out calibration of the no-answer threshold, can improve both overall answer quality and abstention behavior on a laptop GPU.

The system starts from a compact DistilBERT checkpoint trained on SQuAD 1.1. That checkpoint is a deliberately informative baseline: it can extract answer spans, but it has not been trained on adversarial unanswerable questions. The proposed system keeps the architecture fixed, fine-tunes it on SQuAD 2.0, and calibrates the decision boundary on examples that are disjoint from final evaluation.

Planned contributions

- A reproducible preprocessing and training pipeline with sliding context windows and exact character-to-token alignment.
- A controlled comparison separating the contribution of SQuAD 2.0 fine-tuning from post-hoc threshold calibration.
- Answerable-only F1, unanswerable accuracy, and qualitative error analysis in addition to aggregate EM/F1.

Expected outcome

We expect SQuAD 2.0 fine-tuning to produce the largest gain by teaching the model to use the [CLS] position as a no-answer alternative. Calibration should yield a smaller but measurable gain by correcting the operating point for the local evaluation distribution. The project will treat a negative calibration result as informative rather than selecting a threshold on the final test examples.

Research paper summary

Rajpurkar, Jia, and Liang (2018): Know What You Don't Know

Problem and motivation

SQuAD 1.1 contains only answerable questions. A system can therefore improve by always finding the most plausible span, even when its evidence is weak. Rajpurkar et al. expose this limitation by adding more than 50,000 unanswerable questions written adversarially to resemble answerable ones. Success on SQuAD 2.0 requires both reading comprehension and selective prediction: the model must answer when evidence exists and abstain otherwise.

Dataset construction

Crowdworkers were shown a paragraph and an existing answerable question, then asked to write a related question that the paragraph could not answer. This creates hard negatives with high lexical overlap and plausible distractors. The resulting benchmark combines these questions with the original SQuAD examples and evaluates both exact string match and token-level F1.

Main findings

The paper shows a sharp gap between performance on SQuAD 1.1 and SQuAD 2.0. A strong neural reader that attained roughly 86 F1 on the original benchmark fell to about 66 F1 after unanswerable questions were introduced. The result demonstrates that span extraction alone is insufficient: robust systems need an explicit mechanism for comparing the best answer span with a no-answer option.

Connection to this project

The proposed score difference follows the paper's central formulation. For each example, the system compares the [CLS] no-answer score with the highest valid span score. The threshold is not assumed to be zero; it is learned on a held-out calibration set, then frozen for final evaluation.

Methodological context

Pretrained contextual representations and compact deployment

BERT and extractive QA

BERT represents the question and passage jointly with bidirectional self-attention. A small task-specific head predicts start and end logits for each input token. For unanswerable questions, both targets are assigned to [CLS], so the same logits support span extraction and abstention without a separate generative decoder.

Why DistilBERT

DistilBERT compresses BERT through knowledge distillation. It retains much of BERT's language-understanding capability with fewer layers and lower inference cost. This makes it appropriate for a reproducible course project on an 8 GB laptop GPU while still testing the transfer-learning ideas discussed in the final-project handout.

Baseline and ablations

- Baseline A: SQuAD 1.1 checkpoint with the default threshold of zero.
- Ablation B: the same checkpoint with threshold calibration only.
- Model C: SQuAD 2.0 fine-tuning with the default threshold.
- Final system D: SQuAD 2.0 fine-tuning plus held-out threshold calibration.

Evaluation design

The SQuAD 2.0 validation set will be shuffled with a fixed seed and divided into a calibration subset and a non-overlapping final evaluation subset. The project will report EM, token F1, answerable-only EM/F1, and unanswerable accuracy. Qualitative examples will include both improvements and regressions.

Execution plan and risks

Concrete milestones, compute budget, and success criteria

Plan

- Restore the Stanford starter repository and preserve its BiDAF implementation for provenance.
- Implement unit-tested normalization, span scoring, no-answer decisions, and threshold selection.
- Run a tiny end-to-end smoke test before the full experiment.
- Fine-tune for two epochs on 12,000 seeded training examples; use 1,500 examples for calibration and 1,500 for final evaluation.
- Save the exact configuration, software versions, model checkpoint, predictions, figures, and error-analysis cases.

Success criteria

The primary criterion is a clear improvement in held-out F1 over the unadapted SQuAD 1.1 checkpoint. Secondary criteria are increased unanswerable accuracy without collapsing answerable F1, a reproducible run on the available GPU, and a transparent analysis of failure modes.

Risks and mitigations

A subset experiment may not match leaderboard performance; the report will avoid making that claim. Calibration may overfit if tuned on evaluation examples, so the split is fixed before inference and the final evaluation records are never used during threshold selection. Long contexts can truncate answers; a 64-token stride creates overlapping windows. Mixed precision can destabilize optimization; bfloat16 is used when supported, with gradient clipping and deterministic seeds.

Planned deliverables

The final package will include source code, requirements, tests, model and tokenizer files, raw JSON metrics, predictions, training and metric plots, a 6-8 page report, and a research poster.

Key references

Rajpurkar et al. (2018), Devlin et al. (2019), Sanh et al. (2019), and Seo et al. (2017). Full bibliographic entries are provided with the project source.